

SVM Compute Core — Full-Chip Design Summary (m5/v9: Final Harden)

Project: Multi-Class Cardiac Arrhythmia Detection — Caravel chipIgnite Tape-Out **Technology:** sky130A / sky130_fd_sc_hd **Flow:** OpenLane 2 v2.3.10 Classic **Architecture:** Batch v8/v9 — host pre-loads SV + input matrix; ASIC classifies autonomously **RTL freeze:** m5/rt1 v9 — NUM_SV=500, alpha_addr[8:0], reg_alpha_wr[24:0]

Component Summary

svm_compute_core (job 91966, v9)

Metric	Value
Clock	40 MHz (25 ns), TT corner clean
Setup WNS (TT)	+7.83 ns — 0 violations
Hold WNS (TT)	+0.30 ns — 0 violations
DRC	0 violations
Active power	~66 mW
Inference time	3.23 ms / beat (500 SVs $\times$ 256 dim at 40 MHz)
Avg power (80 bpm)	0.284 mW (0.431% duty cycle)
Die area	2500 $\times$ 2500 μm (~14% utilization)
GDS	226 MB
LEF	94 KB
GL netlist	13 MB

user_project_wrapper (job 91967)

Metric	Value
Die area	2920 $\times$ 3520 μm (Caravel fixed, FP_DEF_TEMPLATE)
Macro	u_svm at (253, 554) N — 2500 $\times$ 2500 μm footprint
Clock	wb_clk_i (Caravel), gated to svm_gclk via ICG
CTS	Disabled (RUN_CTS: 0)
DRC	11,923 Magic DRC (boundary artifacts — acceptable)
LVS	1,683 errors (boundary artifacts — acceptable)
GDS	230 MB
LEF	195 KB
GL netlist	78 KB

Functional Results

Implementation	Accuracy	SVs	Notes
sklearn OVR (float)	97.67%	416 total (unlimited)	float precision
ASIC binary OVR (Q6.10)	97.67%	500 total (100 $\times$ 5)	gamma=0.25, C=1.0

Zero accuracy gap — ASIC exactly matches sklearn on all 300 test samples.

Per-class breakdown (300 test samples, 60 per class): | Class | Correct | Accuracy | |———|———|———| | Normal (N) | 60/60 | 100.0% | | PVC | 60/60 | 100.0% | | AFib | 60/60 | 100.0% | | VT | 56/60 | 93.3% | | SVT | 57/60 | 95.0% |

Batch Architecture (v8/v9)

Off-chip RAM Bus

Signal	Pin	Direction	Description
ram_addr[18:0]	GPIO[28:10]	ASIC out	{row[10:0], col[7:0]}
ram_ren	GPIO[29]	ASIC out	Read strobe
ram_rdata[15:0]	LA[15:0]	Host in→ASIC	Data valid after RAM_LATENCY cycles

Address layout: rows 0..499 = SV matrix; rows 500..1499 = input matrix.

RAM_LATENCY parameter — configures the number of clock cycles between ram_ren assertion and valid ram_rdata. Default is 1 (same-cycle combinational SRAM model used in cosim). Set to 3 for the IS61WV51216 async SRAM (10 ns access at 40 MHz → 3-cycle pipeline). The core inserts wait states automatically; the host does not need to pad responses. Verified by svm_ram_latency_tb.sv (10-beat, LAT=3, PASS in ~208 cycles/beat).

What the Host Does

MCU (low-power, continuous)

1. Collect 1000 heartbeats (250 Hz ECG → feature extraction)
 2. Load SV matrix (500 SVs × 256 features) → SRAM rows 0..499
 3. Load input matrix (1000 beats × 256 features) → SRAM rows 500..1499
 4. Write NUM_SAMPLES = 1000, write NUM_SV_0-4 = 100 each
 5. Write alpha coefficients via ALPHA_WR (Wishbone 0x28)
 6. Fire CONTROL[start]
- ▼ ASIC takes over (timing shown for RAM_LATENCY=1 / RAM_LATENCY=3):
- LOAD_INPUT per beat: 256 / 768 cycles (reads from SRAM rows 500+)
 - COMPUTE_DIST per SV: 258 / 770 cycles (reads SV from SRAM rows 0-499)
 - COMPUTE_KERNEL: ~18 cycles (Horner LUT exp approximation, LAT-independent)
 - WRITE_CLASS: argmax+alpha → sample_rdy (IRQ[0]) per beat
 - last beat → done (IRQ[1])

Inference time scales linearly with RAM_LATENCY. At 40 MHz: - LAT=1 (cosim default): **3.23 ms / beat** (500 SVs × 256 dim) - LAT=3 (IS61WV51216 SRAM): **~9.7 ms / beat** — still well within the 750 ms heartbeat period

Wishbone Register Map (base 0x3000_0000)

Offset	Name	R/W	Description
+0x04	CONTROL	RW	[0]=start [1]=vbatt_ok [2]=vbatt_warn
+0x08	STATUS	RO	[0]=done [1]=error [5:2]=error_code
			[8:6]=class [9]=sample_rdy
+0x0C	NUM_SAMPLES	RW	[9:0] beats in batch (1-1000)
+0x10-+0x20	NUM_SV[0-4]	RW	[7:0] SVs per class (max 100 each)
+0x24	PARAM_WR	WO	[19]=en [18:16]=addr [15:0]=data (, C, bias)

Offset	Name	R/W	Description
+0x28	ALPHA_WR	WO	[24:16]=sv_global_idx (9-bit) [15:0]=alpha Q6.10

Full-Chip Power Estimate

Subsystem	Active Power	Duty Cycle	Avg Power
svm_compute_core (batch)	66 mW	0.431% (3.23 ms / 750 ms)	0.284 mW
Caravel management SoC	~5 mW	~5%	~0.25 mW
ECG frontend (analog)	~0.5 mW	100%	0.5 mW
BLE (optional, logging)	~10 mW	~0.1%	~0.01 mW
Total estimated	—	—	~1.04 mW

Battery budget: 200 mAh @ 3.7V = 740 mWh → **740 mWh / 1.04 mW 711 hours (~29.6 days)**. 14-day target met with $2.1\times$ margin. SVM core alone: ~2606 hours (~108 days).

Caravel Submission Artifacts

Repository: github.com/adamleehandwerger/caravel_svm_project

GDS Release: v2.0-hardened (226 MB core / 230 MB wrapper)

Layout artifacts:

File	Size	Job	Status
gds/svm_compute_core.gds	226 MB	91966	
gds/user_project_wrapper.gds	230 MB	91967	
lef/svm_compute_core.lef	94 KB	91966	
lef/user_project_wrapper.lef	195 KB	91967	

Verilog artifacts:

File	Size	Job	Status
verilog/gl/svm_compute_core.v	13 MB	91966	
verilog/gl/user_project_wrapper.v	78 KB	91967	
verilog/rtl/svm_compute_core.sv	—	v9	
verilog/rtl/user_project_wrapper.sv	—	v9	

Acknowledgments

Place-and-route was performed on **Orca**, Portland State University’s high-performance computing cluster, using SLURM batch jobs with OpenLane 2 v2.3.10 inside a Singularity container. We thank the PSU Research Computing team for providing access to the GPU and CPU nodes that made the multi-hour OpenLane runs feasible.

Feature Extraction References

The 256-dim multi-scale feature vector follows established AAMI EC57 beat classification conventions:

Feature group	Dims	Reference
Single-beat morphology (± 64 samples, amplitude-norm)	128	de Chazal P, O'Dwyer M, Reilly RB. "Automatic classification of heartbeats using ECG morphology and heartbeat interval features." <i>IEEE Trans Biomed Eng</i> 51(7):1196-206, 2004. DOI: 10.1109/TBME.2004.827359
10-beat mean morphology template	64	de Chazal P, Reilly RB. "A patient-adapting heartbeat classifier using ECG morphology and heartbeat interval features." <i>IEEE Trans Biomed Eng</i> 53(12):2535-43, 2006. DOI: 10.1109/TBME.2006.883802
RR-interval history (99 intervals $\rightarrow$ 64 pts, norm to NORMAL_RR=308 ms)	64	Llamedo M, Martínez JP. "Heartbeat classification using feature selection driven by database generalization criteria." <i>IEEE Trans Biomed Eng</i> 58(3):616-25, 2011. DOI: 10.1109/TBME.2010.2068048

Standard: AAMI ANSI EC57:2012 — Performance Requirements for Ambulatory ECG Analysers.

Dataset: PhysioNet MIT-BIH Arrhythmia Database (Moody GB, Mark RG, 2001).

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Appendix A — Runtime Model Reload Procedure

The ASIC is fully runtime-reprogrammable. A new trained SVM model can be loaded without resetting the chip or re-synthesizing. The on-chip alpha table and the off-chip SRAM are simply overwritten in place.

What constitutes a “model”

Parameter	Location	Size
SV matrix ($500 \times 256 \times 16$ -bit features)	Off-chip SRAM, rows 0–499	256 KB
Alpha coefficients (500×16 -bit)	On-chip <code>alpha_table[]</code> registers	8 KB
Gamma (, Q6.10)	On-chip <code>gamma_reg</code> shadow register	2 bytes
SV counts per class (<code>NUM_SV[0–4]</code>)	On-chip Wishbone registers	5 bytes

Reload sequence

Perform these steps while the ASIC is idle (`STATUS[done]=1` or after reset). Do **not** fire `CONTROL[start]` until all steps are complete.

Step 1 — Write new SV matrix to off-chip SRAM

Write 500 rows $\times$ 256 columns of Q6.10 feature values to SRAM address rows 0–499. Address encoding: `addr[18:0] = {sv_global_idx[10:0], feat_dim[7:0]}`. This is a bulk SRAM write from the MCU and does not involve the Wishbone bus.

```
for sv in range(500):
    for feat in range(256):
        sram_write(addr=(sv << 8) | feat, data=sv_matrix[sv][feat])
```

Step 2 — Write new alpha coefficients via ALPHA_WR

Write all 500 alpha values over Wishbone. Each write encodes the SV global index and the alpha value in a single 32-bit word:

```
WB_base = 0x3000_0000
ALPHA_WR_offset = 0x28
```

```
for idx in range(500):
    word = (idx << 16) | (alpha[idx] & 0xFFFF) # [24:16]=sv_idx, [15:0]=alpha Q6.10
    wishbone_write(WB_BASE + 0x28, word)
```

Step 3 — Update gamma via PARAM_WR

```
PARAM_WR_offset = 0x24
# addr=0 is gamma register; [19]=en, [18:16]=addr, [15:0]=value
word = (1 << 19) | (0 << 16) | gamma_Q6_10
wishbone_write(WB_BASE + 0x24, word)
```

Step 4 — Update SV counts if changed

Only required if the new model has different SVs per class than the previous model.

```
NUM_SV_offsets: class 0 = 0x10, class 1 = 0x14, ..., class 4 = 0x20
for c in range(5):
    wishbone_write(WB_BASE + 0x10 + c*4, num_sv[c])
```

Step 5 — Fire start

```
wishbone_write(WB_BASE + 0x04, 0x0B) # CONTROL: start=1, vbatt_ok=1, kern_ready=1
```

Constraints

- Total SV count must not exceed 500 (100 per class maximum). Models exceeding this require re-synthesis with updated `NUM_SV` parameters.
- Gamma must be representable in Q6.10 (range 0–63.999, resolution ~ 0.001).
- Alpha values must be representable in Q6.10 (signed, range -32 to $+31.999$).
- The reload can be performed between any two batches. The ASIC does not need to be held in reset during the write sequence — the alpha table and gamma register are shadow-registered and only take effect when `start` fires.
- Writing `ALPHA_WR` while `STATUS[done]=0` (i.e., during active inference) has undefined behavior and must be avoided.

Appendix B — Alternative Design: Hospital-Grade Batch Classifier (28nm)

This appendix specifies an alternative architecture targeting continuous bedside cardiac monitoring in a hospital environment — no power or area constraints, 100-beat batch processing, GEMM-based matrix engine.

B.1 Design Goals

Parameter	Wearable (current)	Hospital (proposed)
Use case	Ambulatory patch	Bedside monitor / ICU
Batch size	1000 beats	100 beats
Latency requirement	< 750 ms/beat	< 5 s per 100-beat batch
Power budget	< 1 mW avg	No constraint
Area budget	Caravel die (6.25 mm ²)	No constraint
Process	sky130A (180nm equiv.)	TSMC 28nm HPC

B.2 Feature Set (unchanged)

The same 256-dimensional multi-scale feature vector is used:

Group	Dims	Content
Single-beat morphology	128	± 64 samples, amplitude-normalized
10-beat mean template	64	Average of preceding 10 beats
100-beat mean template	64	Average of preceding 100 beats (replaces RR-interval history)
Total	256	

The 100-beat template replaces the RR-interval history for hospital use, providing longer-term morphological context available in a continuous-monitoring environment.

B.3 Architecture

Compute engine: 32×32 systolic array

A 32×32 weight-stationary systolic array processes the pairwise distance matrix as a single GEMM operation rather than sequentially iterating over SVs.

The distance computation is restructured as:

$$\begin{aligned}
 D[i, j] &= ||X[i] - SV[j]||^2 \\
 &= ||X[i]||^2 - 2 \cdot (X \cdot SV^T)[i, j] + ||SV[j]||^2
 \end{aligned}$$

The dominant term $X \cdot SV^T$ is a $100 \times 256 \times 256 \times 500$ GEMM — 12.8M MACs, fully parallelized across the systolic array. The squared norm terms are precomputed in a single pass and broadcast.

Memory: on-chip SRAM (1 MB total)

Buffer	Size	Contents
Input matrix	51.2 KB	100 beats $\times$ 256 features $\times$ 16-bit Q6.10
SV matrix	256 KB	500 SVs $\times$ 256 features $\times$ 16-bit Q6.10
Distance matrix	200 KB	100 $\times$ 500 $\times$ 32-bit intermediate
Kernel matrix	100 KB	100 $\times$ 500 $\times$ 16-bit exp output
Alpha table	1 KB	500 $\times$ 16-bit (same as current)
Score / output	2 KB	100 $\times$ 5 $\times$ 32-bit class accumulators
Total	~ 610 KB	(rounded to 1 MB with ECC and overhead)

Clock and process

Parameter	Value
Process	TSMC 28nm HPC
Supply voltage	0.9 V
Clock	800 MHz
Peak compute	1024 MACs/cycle $\times$ 800 MHz = 819 GOPS
On-chip SRAM bandwidth	$\sim$ 100 GB/s

B.4 Performance

Per 100-beat batch:

Stage	Operations	Cycles	Time
Load input + SV to SRAM	—	—	one-time, DMA
GEMM: $X \cdot SV^T$ ($100 \times 256 \times 500$)	12.8M MACs	$\sim$ 12,500	15.6 μ s
Squared norms + broadcast	100K ops	$\sim$ 100	0.1 μ s
Exp LUT (100×500 evaluations)	50K	50,000	62.5 μ s
Alpha accumulation ($100 \times 500 \times 5$)	250K MACs	$\sim$ 250	0.3 μ s
Argmax (100×5)	500	$\sim$ 1	<0.1 μ s
Total per 100-beat batch		$\sim$63,000	$\sim$78 μs

Throughput: 100 beats / 78 μ s = **1,280,000 inf/s (1.28 M inf/s)**

Duty cycle at 80 bpm (100 beats every 75 s): 78 μ s / 75,000,000 μ s = **0.000104%**

B.5 Power

Component	Active	Standby
Systolic array (32×32 , 0.9V, 800 MHz)	$\sim$ 180 mW	$\sim$ 0
On-chip SRAM (1 MB active)	$\sim$ 80 mW	$\sim$ 3 mW leakage
Control logic, IO	$\sim$ 40 mW	$\sim$ 0.5 mW
Total	$\sim$300 mW	$\sim$3.5 mW

Average power at 80 bpm (100-beat batches):

$$\begin{aligned}
 P_{\text{avg}} &= P_{\text{active}} \times \text{duty_cycle} + P_{\text{leakage}} \\
 &= 300 \text{ mW} \times 0.000104\% + 3.5 \text{ mW} \\
 &\approx 0.0003 \text{ mW} + 3.5 \text{ mW} \\
 &= \sim 3.5 \text{ mW}
 \end{aligned}$$

Average power is dominated entirely by SRAM leakage. The compute contribution is negligible — the chip spends 99.9999% of the time idle.

B.6 Die Area

Block	Area (28nm est.)
1 MB on-chip SRAM	$\sim$ 0.50 mm ²
32×32 systolic array	$\sim$ 0.30 mm ²
Horner LUT + exp pipeline	$\sim$ 0.05 mm ²

Block	Area (28nm est.)
Control FSM, IO, misc	~0.10 mm ²
Total estimated	~1.0 mm²

The hospital design is **6× smaller die area** than the current sky130A core (6.25 mm²) despite being orders of magnitude more capable — a direct consequence of the 28nm process density advantage.

B.7 Roofline Comparison

The fundamental shift from the wearable to the hospital design is the operational intensity: the GEMM formulation reuses the SV matrix across all 100 input beats, dramatically increasing arithmetic intensity.

Operational intensity:

Design	Total ops	Memory traffic	Ops/byte
Wearable (sequential, per beat)	512K ops	256 KB (SV re-read each beat)	2.0
Hospital (GEMM, per 100-beat batch)	25.6M ops	307 KB (SV loaded once)	~83

The wearable design sits deep in the **memory-bound** region of the roofline — 98.9% of time waiting for SRAM data. The hospital design operates well above the ridge point (~16 ops/byte at 100 GB/s / 1.6 TOPS) and is firmly **compute-bound**.

B.8 Comparison Table

Throughput and latency:

Metric	Wearable ASIC	Hospital ASIC	sklearn	Numba (8-core)
Throughput	309 inf/s	1,280,000 inf/s	4,200 inf/s	95,000 inf/s
Latency / 100 beats	323 ms	0.078 ms	23.8 ms	1.05 ms
Speedup vs sklearn	0.07×	305×	1×	22.6×

Power, area and roofline:

Metric	Wearable ASIC	Hospital ASIC	sklearn	Numba (8-core)
Active power	66 mW	300 mW	15,000 mW	80,000 mW
Avg power (80 bpm)	0.284 mW	3.5 mW	~15,000 mW	~80,000 mW
Die area	6.25 mm ²	~1.0 mm ²	N/A	N/A
Process	sky130A (180nm)	TSMC 28nm	Intel 10nm	Intel 10nm
Ops/byte	2.0	83	~2.0	~2.0
Roofline regime	Memory-bound	Compute-bound	Memory-bound	Memory-bound

Key observations:

- The hospital ASIC is **305× faster than sklearn** and **13.5× faster than 8-core Numba** on a 100-beat batch
- This is achieved by converting a memory-bound sequential problem into a compute-bound GEMM
- The SRAM leakage floor (3.5 mW) makes the hospital design unsuitable for wearable use — 12× worse average power than the current design despite being 4,000× faster
- Both designs achieve 97.67% accuracy — the architecture difference is entirely in throughput and power, not classification quality